

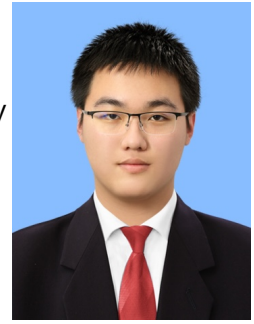
YUWEI ZHAO

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PERSONAL PROFILE

I'm Yuwei ZHAO from Chang Chun, Jilin Province. I'm currently pursuing a dual degree in Computer Science and Technology at Ocean University of China and Heriot-Watt University (UK).

During my studies, I have actively participated in various projects and competitions, gaining extensive hands-on experience—particularly in embedded systems development, computer vision, and intelligent control.

I possess solid programming skills and system implementation experience, and I am proficient with both software and hardware platforms such as STM32, Jetson Orin, RealityKit, and TensorRT. In several research and engineering projects, I served as a core developer, demonstrating strong teamwork and problem-solving abilities.

In the future, I aspire to further my research in the integration of Artificial Intelligence and Robotics —Embodied Intelligence, aiming to advance technological innovation and real-world applications in this emerging field.

EDUCATIONAL BACKGROUND

- **Ocean University of China** August 2023 – June 2026
Bachelor of Computer Science and Technology
 - **Heriot-Watt University** September 2026 – June 2027
Bachelor of Engineering
- [Core Courses] Linux & C (96/100), Robotics Group Project (95/100), Linear Algebra (95/100), Software Development (90.4/100), Discrete Mathematics (88/100)

RESEARCH AND PROJECTS EXPERIENCE

- **Research Experience**
 - **OverLoCK Feature Abstraction and Multi-source Remote Sensing Image Classification Research** · 2025-2026 · Running · Supervisor: Feng GAO [OUC]
- **Robotics Group Project I Teaching Assistant (TA for Robotics Group Project I)** · 2025 · Supervisor: Shu ZHANG [OUC]
 - Assisted in grading over 90 students' assignments and lab reports, designed experimental procedures and testing environments.
 - Explained Python syntax and Robomaster SDK usage in lab sessions, helping students troubleshoot programming and hardware issues.
 - Improved the average project completion rate by 20%.
- **Projects Experience**
 - **Achieve Python 3.8+ Version Compatibility based on Robomaster-SDK** · 2025 · [Github @Robomaster-SDK-Ultra](#)
 - **Rewriting Path Tracing in Swift to Replace Traditional Rasterization for Black Hole Simulation Experiments** · 2025 · [GitHub @BlackHole_Simulator](#)
 - **Offshore Photovoltaic Monitoring and Operation System** · 2025 · System construction and signal transmission; web intelligent control platform construction · In progress

- **Multi-modal Fusion: All-weather Offshore Intelligent Perception** · 2025 · Using monocular depth algorithms to generate radar point cloud images to supplement the dataset · [GitHub @VesselContest_C3](#)
- **Swift Native iOS AR App Development** · 2025 · Implementing arbitrary spatial anchor setting and navigation technology based on RealityKit and ARKit · [GitHub @Macintustin](#)
- **Porting Nim Language to LoongArch64 Architecture** · 2024 · Implementing Nim's environmental adaptation on LoongArch64 architecture based on LLVM backend · [GitHub @Nim2LoongArch64](#)

HONORS AND AWARDS

- 2025 · **China Marine Vehicle Design and Construction Contest C3** · Second Prize of the National · Responsible for generating radar point cloud images using monocular depth algorithms to supplement the dataset
- 2025 · **The Interdisciplinary Contest in Modeling** · Honorable Mention · Responsible for mathematical model construction and data mining
- 2025 · **China Collegiate Computing Contest - Mobile Application Innovation Contest** · Second Prize of the East China Region · Responsible for the full process development of Swift native iOS App
- 2024 · **Computer System Development Capability Competition** · Excellence Award of the East China Region · Responsible for Nim's environmental adaptation on LoongArch64
- 2025 · **China Marine Vehicle Design and Construction Contest F1** · Third Prize of the Shandong Region · Responsible for STM hardware construction and algorithm implementation
- 2024 · **Apple Swift Certification Contest** · Certified User
- 2024-2025 · **Ocean University of China Scholarship** · Third-Class Scholarship (9 / 48)
- 2023-2024 · **Ocean University of China Scholarship** · Third-Class Scholarship (12 / 47)

TECHNICAL SKILLS AND TOOLS

- **Programming Languages:** C/C++, Java, Python, Swift, HTML/CSS, JavaScript, Nim
- **Frameworks and Libraries:** RealityKit, ARKit, OpenCV, YOLOv8, STM32 HAL, TensorRT, PyTorch
- **Embedded Systems:** STM32, Arduino, ESP32, Jetson Orin, PWM control, TIM interrupt, sensor integration
- **Development Platforms:** Linux(Ubuntu), macOS, CUDA GPU acceleration, ROS2(basic)
- **Development Tools:** Git, Docker, Xcode, VSCode, CMake, LaTeX, Markdown
- **Languages:** English(CET-6, 520), German(basic communication)